

#Overview

This project implements a **multimodal regression** approach for real estate price prediction by combining structured housing attributes with satellite imagery to capture both economic and environmental determinants of property value. **Tabular features** represent intrinsic property characteristics such as **size, quality, and location**, while **satellite images** provide neighborhood-level visual context including **green cover, built-up density, and road connectivity**.

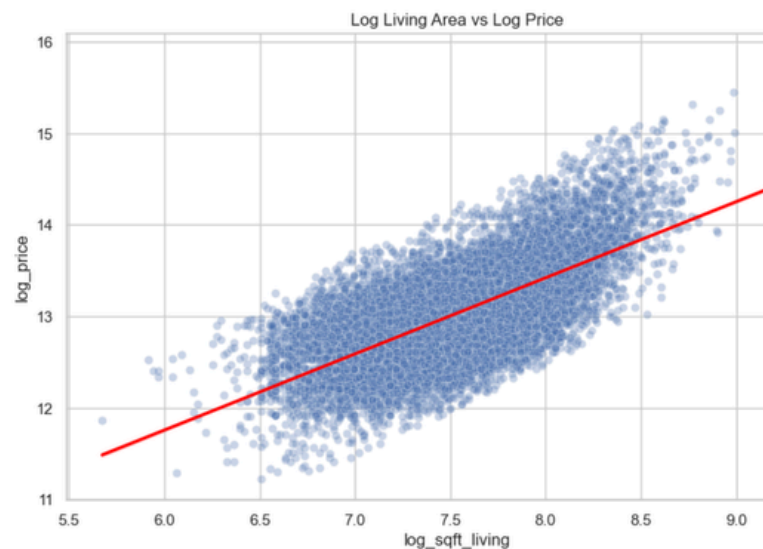
Following initial **data cleaning** and basic **preprocessing**, extensive **exploratory and geospatial analysis** was conducted to study price distributions, spatial patterns, and relationships between property value and both tabular and visual features. These insights guided the final **feature selection** and modeling decisions.

For modeling, multiple tabular-only baseline models were trained to establish performance benchmarks. To incorporate visual information, satellite images were programmatically fetched using latitude and longitude coordinates and processed using a pre-trained **ResNet50** Convolutional Neural Network to extract high-level visual **embeddings**. These embeddings were **fused with tabular features** to train a multimodal regression model, enabling the system to learn complementary signals from both data modalities.

The best-performing model was optimized through **hyperparameter tuning**. Finally, **Grad-CAM-based explainability** was applied to validate that the image model focuses on economically meaningful regions of satellite imagery. Model performance was evaluated by comparing tabular-only and multimodal results, demonstrating the contribution of visual context to improved valuation accuracy.

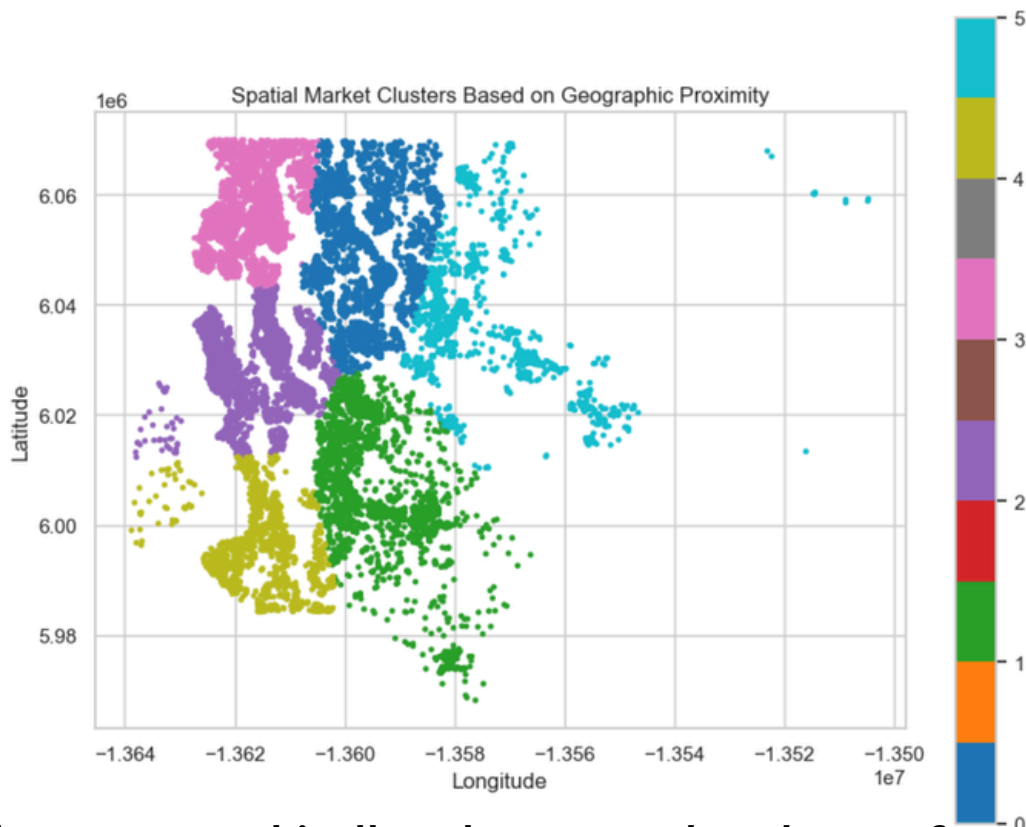
Exploratory Data Analysis

Figure 1: Log of Living Area vs. Log-Transformed Property Price



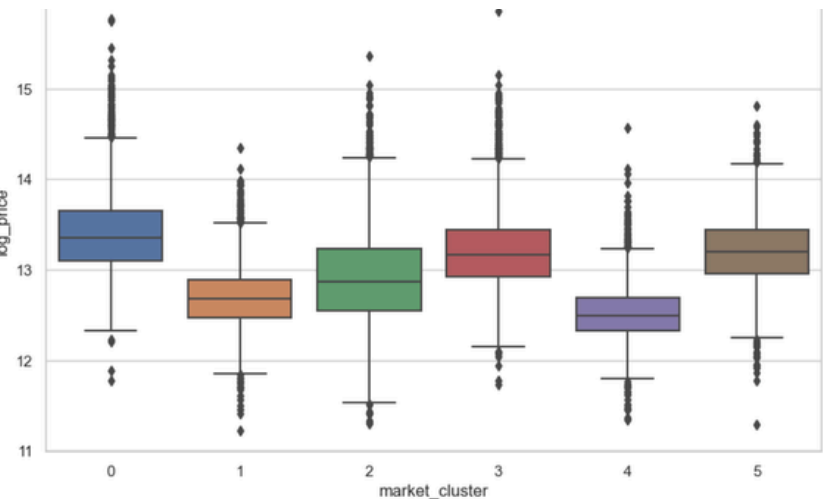
A strong linear relationship is observed between log living area and log price, confirming property size as the primary driver of valuation. The clear trend justifies the use of `log_sqft_living` as a key feature. The remaining dispersion around the regression line represents price variation not explained by size alone, motivating residual modeling and the inclusion of satellite imagery to capture neighborhood-level premiums.

Figure 2: Spatial Market Clusters Based on Geographic Proximity



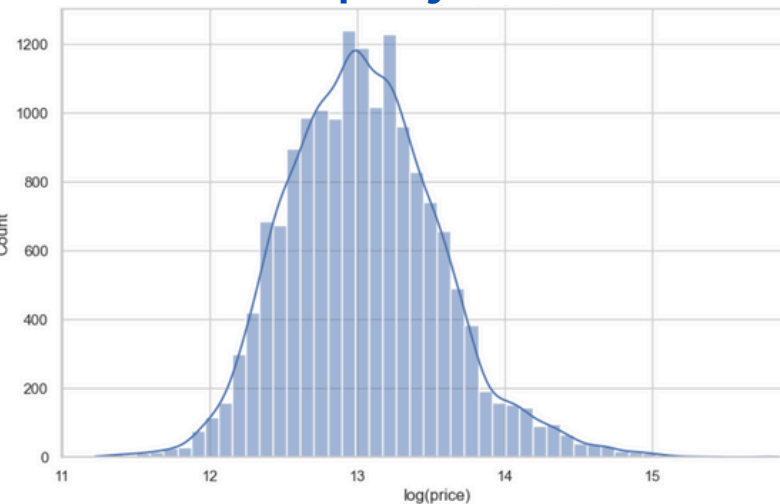
This plot shows geographically coherent market clusters formed using latitude and longitude. Nearby properties group into localized sub-markets with similar pricing behavior, and average prices vary clearly across clusters, highlighting the strong role of spatial segmentation in capturing local market dynamics.

Figure 3: Price Distribution Across Spatial Market Clusters



This boxplot shows clear differences in **price distributions across spatial market clusters**. Clusters exhibit distinct median prices and variability, confirming that **location-driven sub-markets have significantly different valuation patterns**, which justifies incorporating spatial clustering into the modeling pipeline.

Figure 4: Distribution of Log-Transformed Property Prices



Property prices are **highly right-skewed**. Applying a log transformation **reduces skewness, stabilizes variance**, and improves linearity, leading to **better model convergence** and more reliable regression performance. Hence, **log(price) was used as the target variable**.

Figure 5: Bedrooms vs. Log-Transformed Property Price

Property **prices generally increase with the number of bedrooms**, indicating a positive relationship between house size and value. However, the trend shows **diminishing returns beyond a certain point**, where additional bedrooms contribute less to price appreciation. This observation **motivated the use of normalized features such as living area per bedroom** to better capture space efficiency in modeling.

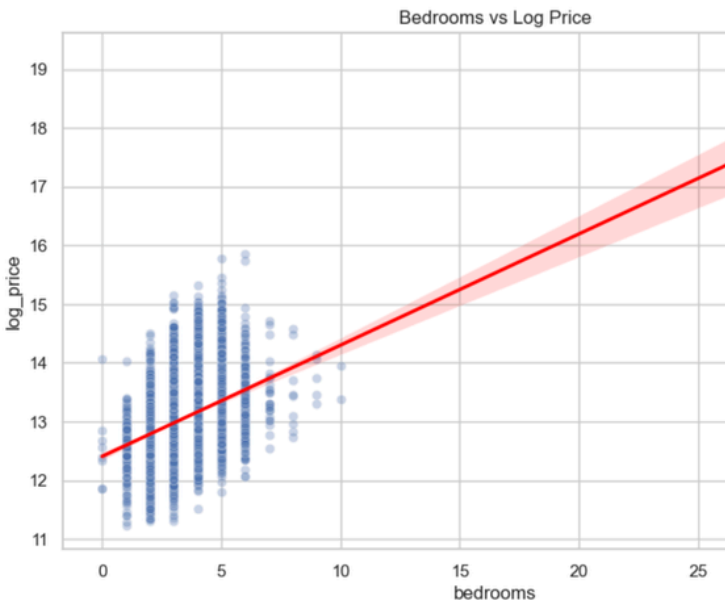
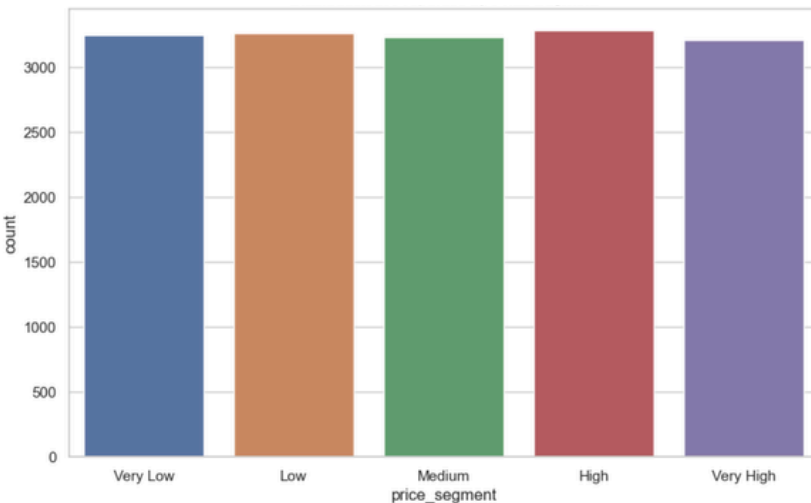
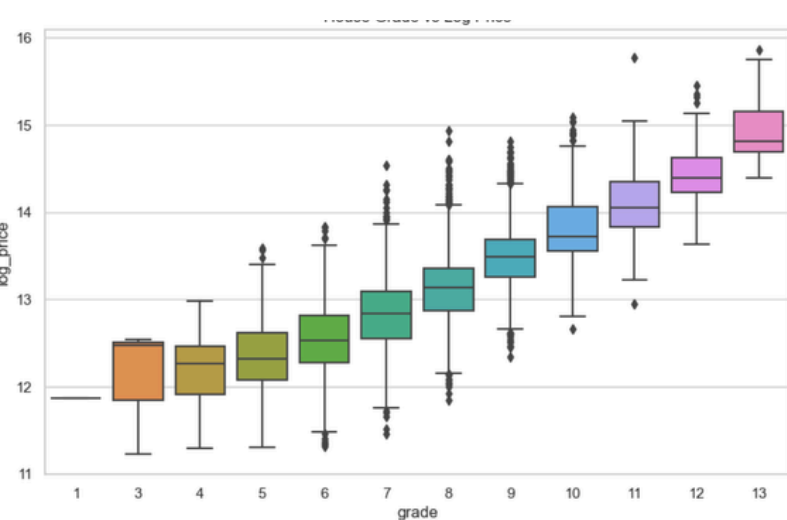


Figure 6: Distribution of Properties Across Price Segments



This bar chart shows that **properties are evenly distributed across the five price segments (Very Low to Very High)**. This balance indicates **minimal price-segment bias, allowing the model to learn consistently** across low-, mid-, and high-value properties and enabling stable training and fair evaluation across price ranges.

Figure 7: House Grade vs. Log-Transformed Property Price



Higher-grade properties consistently command **higher prices**, indicating that **construction quality** and **architectural design** are key value drivers. The **sharp, non-linear price** increases across grade levels suggest threshold effects, **reinforcing the need for non-linear modeling approaches**.

Renovated properties consistently **command higher prices** than non-renovated ones, indicating a clear renovation premium. Representing **renovation status as a binary flag** captures this effect **more robustly than raw renovation year values** and improves model interpretability.

Figure 8: Impact of Renovation on Log-Transformed Property Price

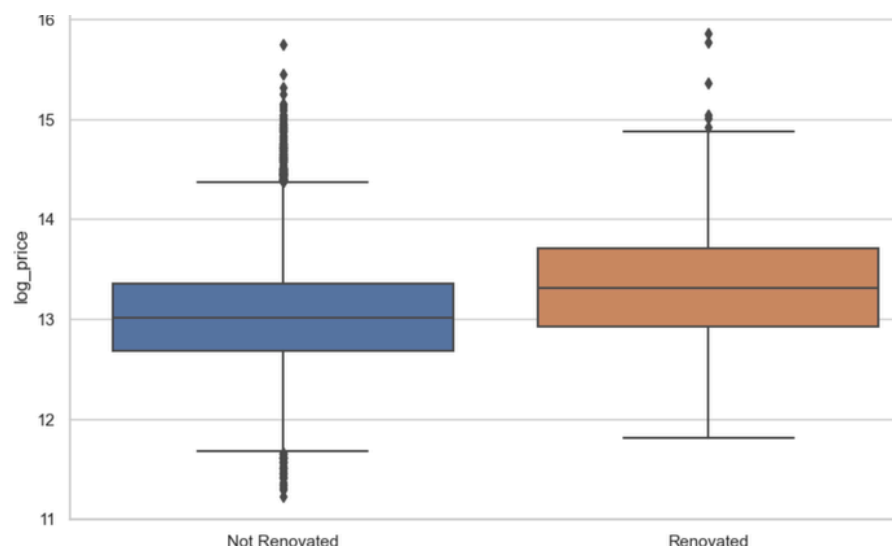
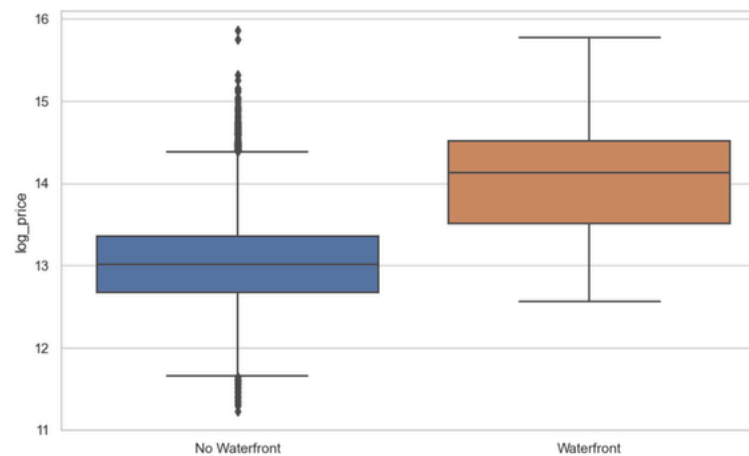


Figure 9: Waterfront Effect on Property Price



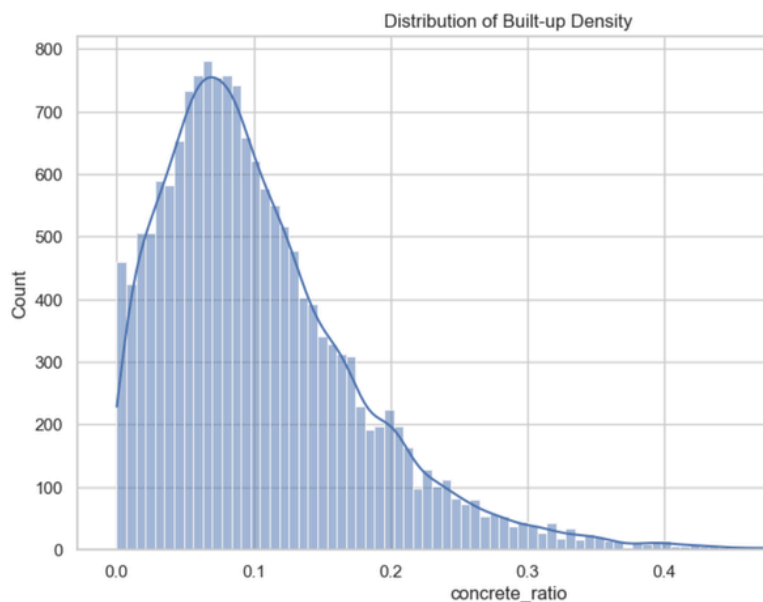
Waterfront properties exhibit a clear **price premium** over non-waterfront homes, highlighting the strong value of natural amenities. This effect represents a **visual signal that satellite imagery can naturally capture** and reinforce in the multimodal model.

Figure 10: Space Efficiency (Living Area per Bedroom) vs. Property Price



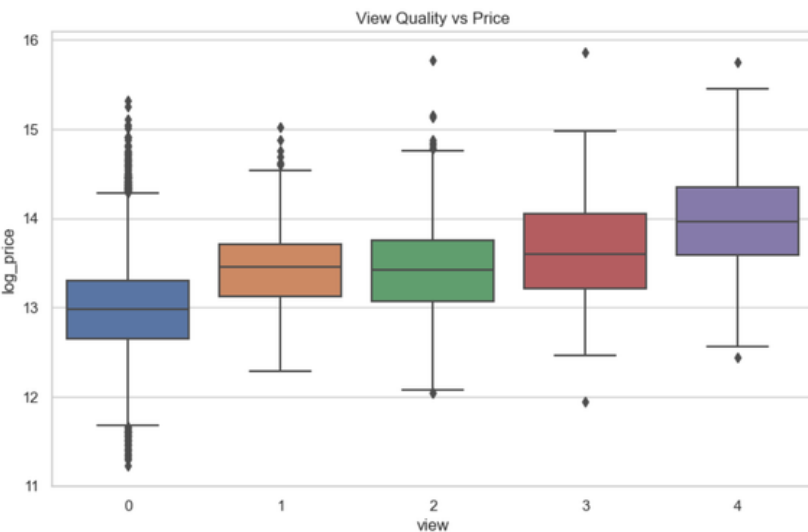
Property prices increase with higher living space per bedroom, indicating that space efficiency is an important value driver beyond bedroom count alone. **Homes offering larger, less crowded bedrooms tend to command higher prices.** This supports the inclusion of the **engineered sqft_per_bed feature to capture layout quality** and effective space utilization in the model.

Figure 11: Distribution of Built-up Density (Concrete Ratio)



This plot shows the distribution of **built-up density (concrete_ratio)** across properties. The **distribution is right-skewed**, indicating that **most neighborhoods have low to moderate built-up density**, while highly concrete-dense areas are relatively rare. **This makes built-up density an informative visual feature for capturing neighborhood characteristics** that influence property prices.

Figure 12: View Quality vs Property Price



This boxplot shows a **clear positive relationship between view quality and property price**. As the **view rating increases, the median log price consistently rises**, indicating that **better scenic views command a strong price premium**. This highlights **view quality as an important value driver** that complements structural and spatial features in pricing models.

Figure 13: Basement Impact on Property Price

Properties with **basements** show a **consistently higher median log price compared to those without basements**. This indicates that basement space adds measurable value, **acting as a price premium due to increased usable area and flexibility**. The clear separation supports using a **binary basement indicator as an effective feature** in the pricing model.

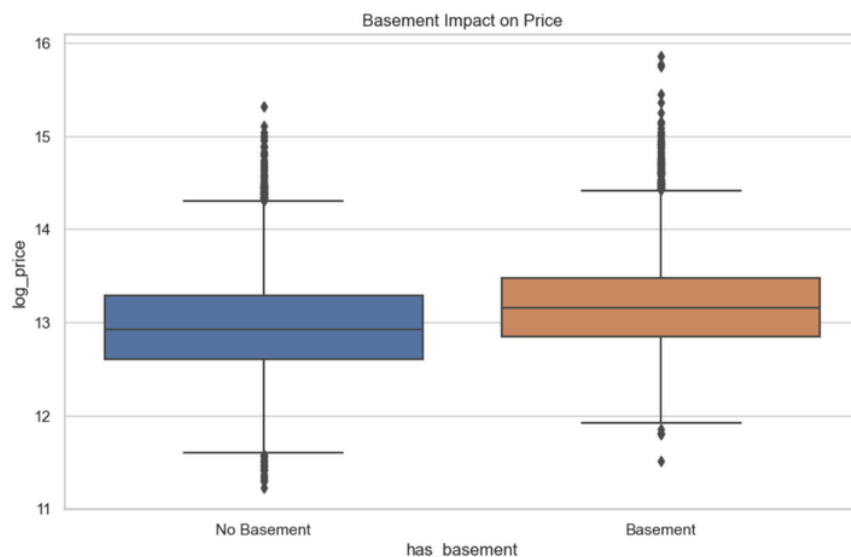
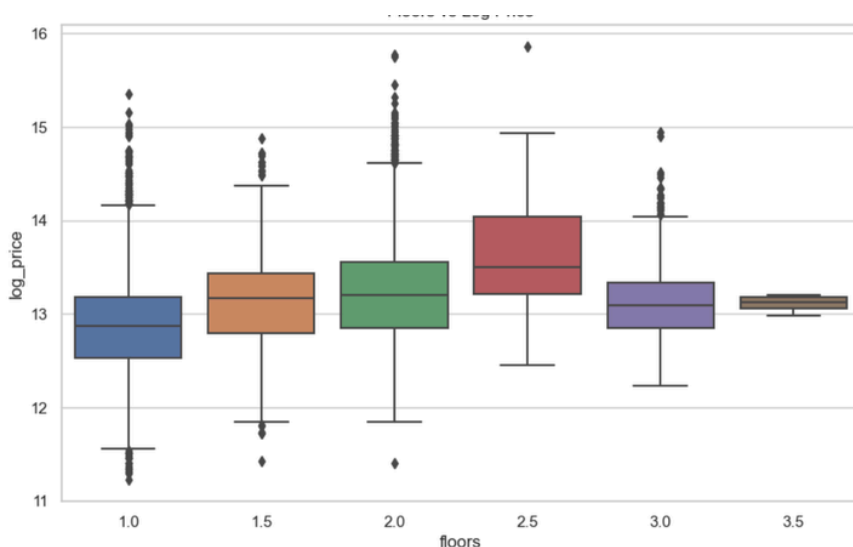


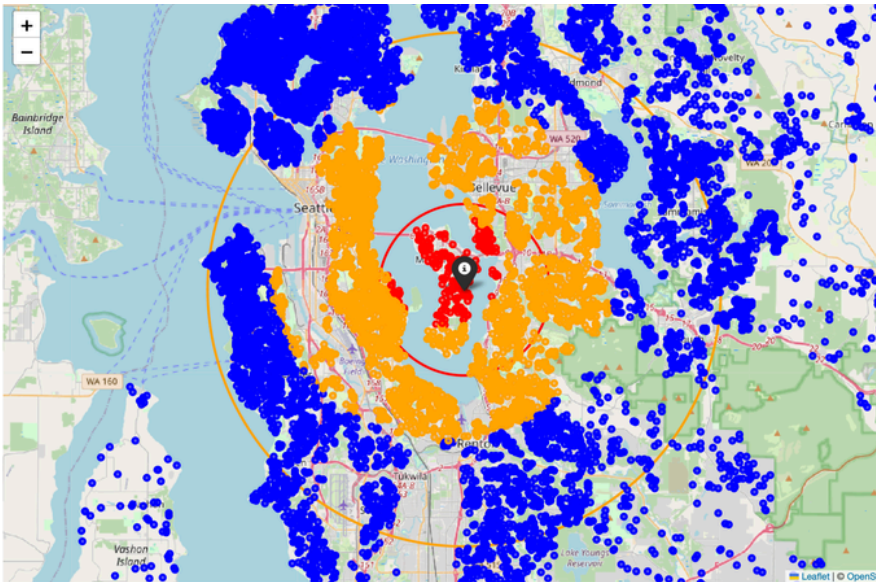
Figure 14: Floors vs Property Price



Property prices **generally increase with floors**, with **2–2.5 floor homes showing the highest median prices**. **Single-floor houses are cheaper**, while very high floor counts show diminishing gains due to rarity. This highlights **floors as a non-linear pricing feature** that **must be modeled carefully**.

Visual-Financial Insights

Figure 15: Geofenced Zones and Prime Location Effect



The **folium-based geospatial visualization** applies **geofencing** to **segment properties** into **prime, urban, and suburban** zones. Properties located within the central **prime zone** generally **exhibit higher prices**, motivating the creation of a **binary prime_zone** feature to **explicitly capture** the **location premium** in the model.

Spatial Distribution & Distance-from-Center Effects on Property Price

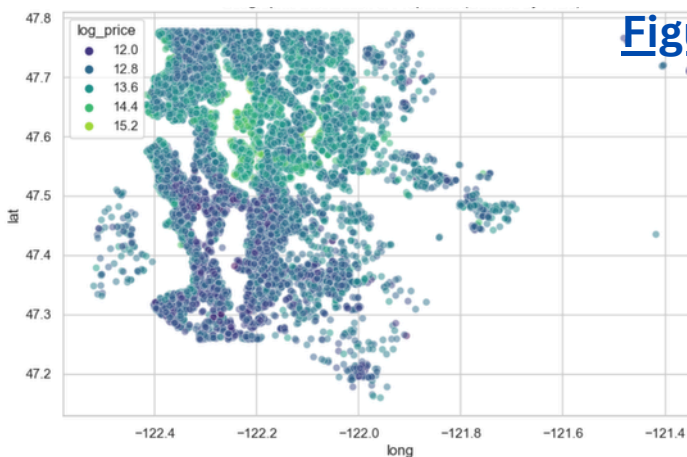
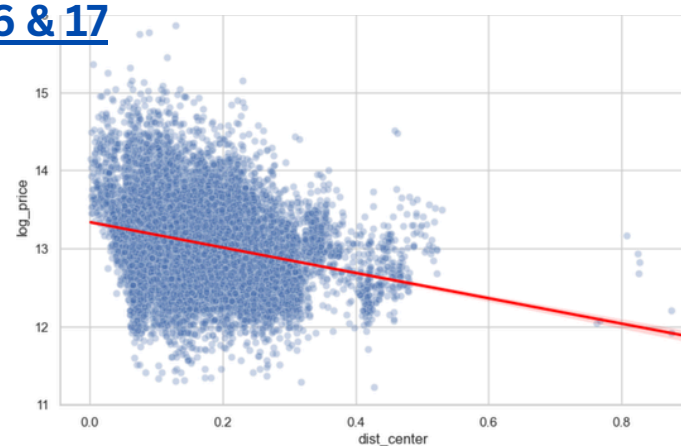


Figure 16 & 17



The geographic distribution of properties colored by price reveals clear **spatial clustering**, with **higher-priced properties concentrated near the city center** and lower-priced properties located toward the periphery. This pattern is **reinforced by the distance-from-center analysis**, which shows a consistent decline in property prices as distance from the urban core increases, reflecting the urban-suburban price premium. These complementary **spatial insights justify the use of satellite imagery** to capture neighborhood context and **motivated the engineering of a dist_center feature**, enabling the model to explicitly learn location-driven valuation effects.

High-Value Property Neighborhoods

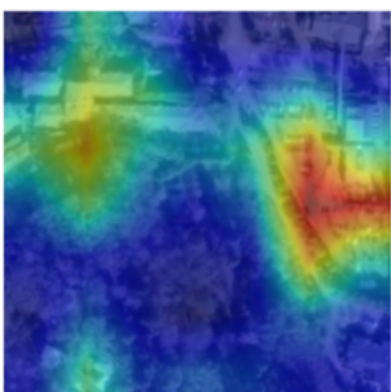


Low-Value Property Neighborhoods



This **visual comparison highlights systematic differences between high-value and low-value property neighborhoods** using satellite imagery. **High-value areas tend to exhibit better-planned layouts, wider roads, and more consistent green cover, reflecting higher neighborhood quality and livability.** In contrast, **low-value neighborhoods show denser or less organized layouts with relatively weaker visual appeal.** These differences demonstrate that neighborhood-level visual characteristics contain meaningful information for property valuation, **justifying the use of CNN-based image embeddings to capture environmental and contextual factors** beyond tabular features.

Grad-CAM Explanation of CNN-Based Price Prediction



ORIGINAL

Grad-CAM

The **Grad-CAM visualization highlights the regions of the satellite image that most influenced the model's price prediction.** In this example, the model **focuses strongly on areas with dense green cover and well-defined road connectivity, as indicated by the high-intensity (warm-colored) regions in the heatmap.** These visual cues correspond to **environmentally desirable neighborhoods**, which are typically associated with higher property values due to improved livability and aesthetics. The **absence of strong activation over irrelevant background regions indicates** that the **CNN learns meaningful neighborhood-level features rather than noise.** This confirms that **satellite imagery contributes interpretable and economically relevant information** to the multimodal valuation model.

#Feature Engineering Summary

Based on insights from detailed exploratory, spatial, and visual analysis, targeted feature engineering was performed to capture key structural, spatial, and temporal drivers of property prices while reducing noise and redundancy.

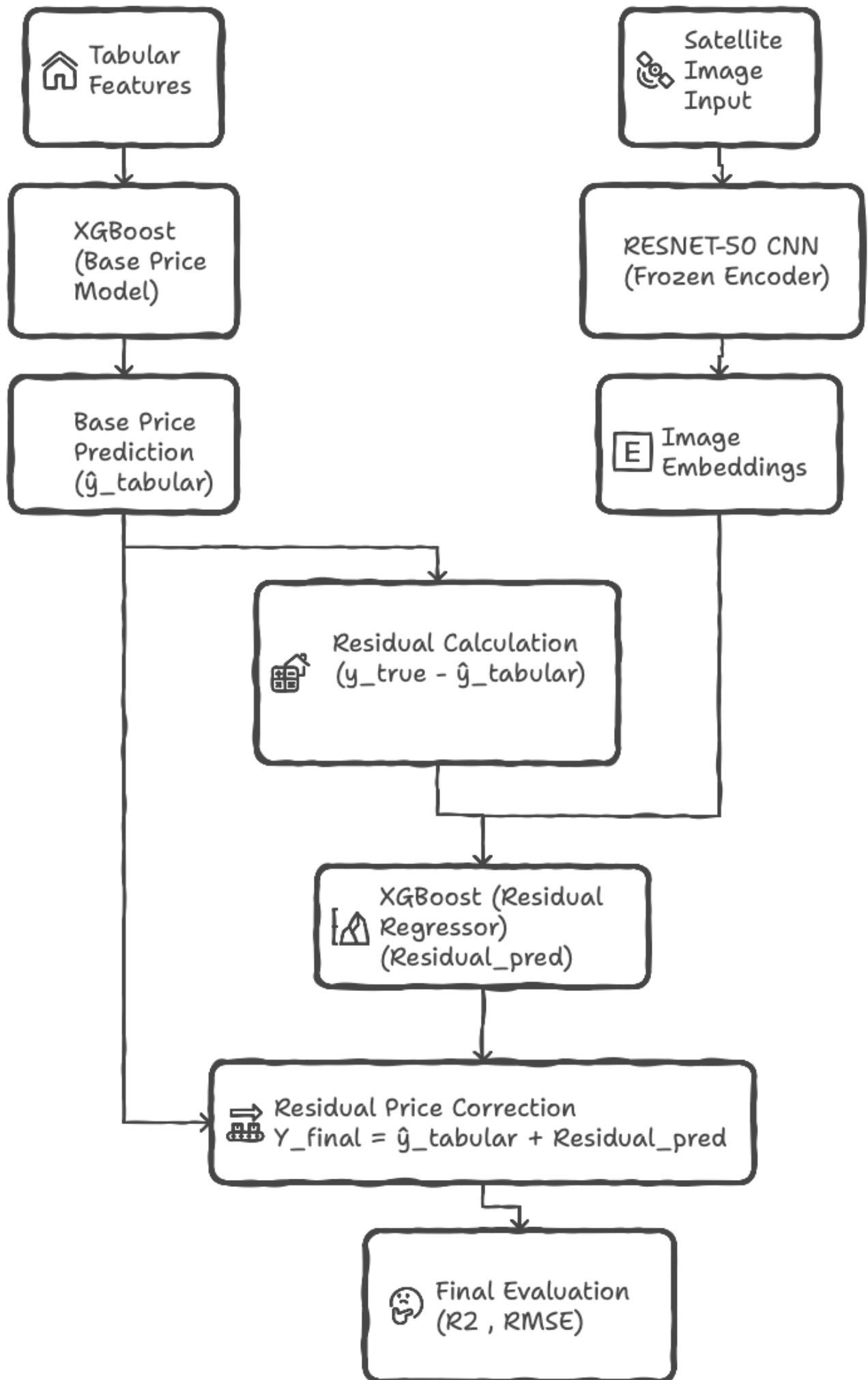
- **Removal of non-informative identifiers:** Dropped **id** and **zipcode** to prevent data leakage and overfitting.
- **Temporal features:** Extracted **year_sold** and **month_sold** from the transaction date to capture time-related pricing patterns.
- **Renovation indicator:** Converted renovation year into a **binary renovated** flag to robustly capture renovation-related price premiums.
- **Property age:** Engineered **house_age** to reflect lifecycle effects on valuation.
- **Log transformations:** Applied log scaling to **sqft_living** (**log_sqft_living**) and target variable (**log_price**) to **reduce skewness** and stabilize variance.
- **Space-efficiency ratios:** Created **sqft_per_bed** and **bath_per_bed** to capture layout quality beyond raw size measures.
- **Structural composition:** Introduced **above_ratio** to represent the proportion of above-ground living space.
- **Spatial features:** Engineered **dist_center** to model urban-suburban price gradients and **lat_long** as a geographic interaction term.
- **Feature pruning:** Removed redundant raw features (e.g., original square footage and construction year) to reduce multicollinearity.
- **Indexing:** Added **row_id** for tracking and reproducibility.

#Experimental Setup

The following setup outlines the **data preparation, feature processing, and modeling choices** used to evaluate both tabular and multimodal approaches in a consistent and controlled manner.

- The data was split into **70% training, 15% validation, and 15% testing** for reliable performance evaluation.
- Tabular features were **standardized using StandardScaler**, fitted on the training set and applied to validation and test sets.
- **Satellite images were encoded using a pre-trained ResNet50**, producing image embeddings that were **reduced from 2048 to 512 dimensions using PCA**.
- **XGBoost** was used as the **primary regression model for both tabular prediction and image-based residual learning**.

#Architecture Diagram of the Residual Multimodal Price Prediction Model



#Results

Tabular **vs.** Tabular + Satellite Images

Model	R ² (Test)	RMSE (log_price) (Test)
Linear Regression (Tabular only)	0.7855	0.2465
Neural Network (Tabular only)	0.8798	0.1845
XGBoost (Tabular only)	0.9084	0.1611
XGBoost (Residual Multimodal)	0.9096	0.1600

#Discussion: Impact of Satellite Imagery on Model Performance

The tabular XGBoost model achieves strong baseline performance ($R^2 = 0.9084$), indicating that structured housing attributes already explain around 90% of the variance in property prices. Incorporating satellite imagery through residual learning improves performance to $R^2 = 0.9096$ with a reduced RMSE ($0.1611 \rightarrow 0.1600$). While the numerical gain is modest, it is meaningful at this accuracy level, showing that satellite images contribute additional neighborhood-level and environmental information—such as greenery, urban density, and spatial layout—beyond structured data. These gains reflect the model’s ability to capture subtle price premiums not explained by tabular features alone.

#Why Satellite Images Were Integrated Only with XGBoost?

Satellite images were integrated only with XGBoost because it achieved the strongest performance on tabular data. Using a weaker baseline could obscure the true contribution of visual features. XGBoost therefore provides a fair and interpretable foundation to evaluate the added value of satellite imagery.

#Conclusion

This project demonstrates the effectiveness of a **multimodal residual learning framework** for real estate price prediction by integrating structured housing data with satellite imagery. **Tabular features alone were able to explain nearly 90% of the variance in property prices**, highlighting their strong explanatory power. The inclusion of satellite imagery through **CNN-based embeddings provided a small but consistent performance improvement**, capturing neighborhood-level and environmental effects not fully represented in tabular data. **Grad-CAM-based explainability** further confirmed that the **image model focuses on economically meaningful visual features such as greenery, road structure, and spatial layout**. Overall, the results validate the use of satellite imagery as a refinement mechanism for high-performing tabular models.

#Future Work

- Incorporating **higher-resolution or multi-scale satellite imagery** to capture finer neighborhood details.
- Extending the framework to **temporal satellite data** to model changes in urban development over time.
- Exploring **end-to-end multimodal training** instead of fixed CNN embeddings.
- Integrating additional contextual data sources such as **points of interest (POIs), socioeconomic indicators, or traffic patterns** to further enhance valuation accuracy.

#References

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https://services.arcgisonline.com/ArcGIS/rest/services/World_Imagery/MapServer

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3.Grad-CAM: Visual Explanations for CNNs

<https://arxiv.org/abs/1610.02391>

4.House Price Prediction using Satellite Imagery (paper)

<https://arxiv.org/pdf/2105.06060>